Typhoid Fever

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Introduction to Typhoid Fever

- Bacterial infection caused by *Salmonella enterica* serovar Typhi.
- Global burden: ~11–21 million cases/year, ~150,000 deaths.
- Common in areas with poor sanitation and limited access to clean water.
- Significant public health challenge in South Asia, sub-Saharan Africa, and parts of Latin America.

Causative Organism & Virulence Factors

- **Salmonella Typhi**: Gram-negative, motile, facultative intracellular bacillus.
  - **Vi antigen**: capsular polysaccharide, inhibits opsonization and phagocytosis.
  - **Endotoxin (LPS)**: causes systemic inflammation and fever.
  - **Pathogenicity Islands**: SPI-1 (epithelial invasion), SPI-2 (macrophage survival).
  - **Type III Secretion System (TTSS)**: injects effector proteins into host cells

Transmission & Risk Factors

Fecal-oral route: ingestion of contaminated food or water.

Human reservoir only — asymptomatic carriers play a major role.

Chronic carriage common in the gallbladder.

Risk factors: poor hygiene, urban slums, travelers, contaminated produce/dairy.

Pathogenesis

Bacteria survive gastric acid → penetrate ileal M cells → transcytosis.

Phagocytosed by macrophages, survive intracellularly.

Disseminate via lymphatic system and bloodstream (primary bacteremia).

Invade liver, spleen, bone marrow → replicate → secondary bacteremia.

Peyer's patches in ileum become inflamed → necrosis and potential perforation.

Incubation and Disease Progression

Incubation: 6 to 30 days (avg ~10–14 days).

Week 1: Gradual onset, fever, malaise, headache, dry cough.

Week 2: Higher fever, abdominal pain, rose spots, bradycardia.

Week 3: Typhoid state — toxic appearance, confusion, risk of complications.

Week 4: Either recovery begins or complications worsen.

Clinical Features

- Step-ladder pattern fever, reaching 39–40°C.
- **Gastrointestinal**: constipation (early), diarrhea (later), RLQ pain.
- **Neurological**: delirium, stupor (“typhoid state”), irritability.
- **Others**: hepatosplenomegaly, rose spots on abdomen, coated tongue, bradycardia.

Complications

- **assignment**

Investigations & Diagnosis

Blood culture: most sensitive in Week 1.

Stool/urine culture: better yield in Week 2–3.

Bone marrow culture: gold standard, highest sensitivity even after antibiotics.

Widal test: detects O/H antibodies, low specificity

CBC: leukopenia, anemia, thrombocytopenia.

LFTs may show mild transaminitis.

Antibiotic Treatment (Doses & Duration)

Uncomplicated Typhoid:

- *Azithromycin*: 500 mg once daily for 7 days.
- *Cefixime*: 400 mg BID for 7–14 days.

Severe/complicated Typhoid:

- *Ceftriaxone*: 1–2 g IV daily for 10–14 days.
- *Meropenem*: for XDR cases.

Antibiotic Treatment (Doses & Duration)

- **Alternatives** (with susceptibility):
  - Fluoroquinolones (e.g., ciprofloxacin) – 500 mg BID × 7–10 days.
  - Not recommended in resistant areas.

Supportive Care & Resistance Monitoring

- IV fluids for dehydration.
- Antipyretics for fever.
- Nutritional support, electrolyte correction.
- Resistance is rising: MDR (multi-drug resistant), XDR (extensively drug-resistant) strains.
- Importance of culture and sensitivity testing before choosing antibiotics.

Prevention & Vaccination

- Improved sanitation, clean water, food hygiene.
- Identify and treat carriers.
- **Vaccines**:
  - *Vi polysaccharide* (IM, ≥2 years old).
  - *Ty21a* oral live attenuated (≥6 years).
  - *Conjugate vaccines* (newer, longer-lasting, for infants).
- Public health measures: surveillance, education, travel advisories.

Conclusion



Typhoid remains a serious but preventable disease.



Early diagnosis and treatment prevent complications and mortality.



Drug resistance is a growing threat — vaccination and public health strategies are key.



Continued efforts needed for eradication, especially in endemic areas.

Thank you